Stainless steel (stain -free iron) is an alloy whose constituent elements generally are approximately as follows :

Iron	–	89.4%
Chromium	–	10.0%
Manganese	–	0.35%
Carbon	–	0.25 %

Silicon is not the constituent element of stainless steel, while few quantity of nickel is also added as a constituent element in stainless steel.

27. The important metal used with iron to produce stainless steel, is :

(a) Aluminium (b) Chromium
(c) Tin (d) Carbon

U.P.P.C.S. (Pre) 2002

Ans. (b)

See the explanation of above question.

28. Stainless Steel is an alloy in which following is added along with iron:

(a) Zinc (b) Chromium
(c) Tin (d) Copper

U.P. R.O./A.R.O. (Mains) 2017

Ans. (b)

See the explanation of above question.

29. Which of the following elements are included in stainless steel?

(a) Chromium, Zinc, Carbon and Iron
(b) Nickel, Iron, Zinc and Tin
(c) Iron, Chromium, Manganese and Carbon
(d) Iron, Zinc, Manganese and Tin

U.P.P.S.C. (GIC) 2010
Chhattisgarh P.C.S. (Pre) 2008

Ans. (c)

See the explanation of above question.

30. What is mixed with iron to make stainless steel?

(a) Nickel and Copper (b) Zinc and Tin
(c) Nickel and Tin (d) Chromium and Nickel

Uttarakhand P.C.S. (Pre) 2005
R.A.S./R.T.S. (Pre) 1996

Ans. (d)

See the explanation of above question.

31. Stainless steel is an alloy of :

(a) Iron and nickel
(b) Iron and chromium
(c) Copper and chromium
(d) Iron and zinc
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

See the explanation of above question.

32. How much carbon does steel contain?

(a) 0.1–2% (b) 7–10%
(c) 10–50% (d) Zero

M.P.P.C.S. (Pre) 2000

Ans. (a)

The amount of Carbon contains in Steel lies between 0.1% to 2%.

Non-Metals

A. Carbon and its Various Forms

Notes

- The common properties of non-metals are –

(i) High ionization energies.
(ii) High electronegativities.
(iii) Poor thermal conductors.
(iv) Poor electrical conductors.
(v) Brittle solids-not malleable or ductile.
(vi) Little or no metallic luster.
(vii) Gain electrons easily.
(viii) Dull, not metallic-shiny, although they may be colourful.
(ix) These are generally gases as - Hydrogen, Oxygen, Fluorine, Chlorine, Argon, Krypton, etc.; solids as - Carbon, Phosphorus, Sulphar, Selenium, Iodine, etc.; and liquid - Bromine (only).

Oxocarbon :

- An oxocarbon or oxides of carbon is an inorganic compound, consisting only of **carbon** and **oxygen.**
- The simplest and most common oxocarbons are **carbon monoxide** and **carbon dioxide** with IUPAC (International Union of Pure and Applied Chemistry) names **Carbon (II) oxide** and **Carbon (IV) oxide** respectively.

Carbon has three main allotropes :

(a) Graphite

(b) Diamond

(c) Buckminsterfullerene

Graphite :

- Graphite archaically referred to as **plumbago -** The stick that left a mark. Lead pencils have always been made of graphite. The mineral was so much like the lead ores found at that time that the residents called it plumbago - which is Latin for lead ore.
- It is a crystalline form of the element carbon with its atoms arranged in a **hexagonal** structure.
- It occurs naturally in this form and is the most stable form of carbon under standard conditions.
- Under very high pressure and temperature, it converts into a diamond.
- It is soft and used to prepare **lead pencils**.
- This is useful in applications where wet lubricants, such as oil can not be used. So it is called **dry lubricant**.
- It is a good conductor of electricity.

Diamond :

- Diamond is a solid form of carbon with its atoms arranged in a crystal structure called **diamond cubic**.
- Diamonds are used as an abrasive because it is very hard. Small particles of diamonds are embedded in saw blades, drill bits and grinding wheels for the purpose of cutting, drilling or grinding hard materials.
- It has a high index of refraction and high luster. Due to these properties, diamond is the most valuable and popular gemstone of the world.
- Its thermal conductivity is highest with respect to other natural materials.
- It is used to inscribe words on glass.
- Hydrofluoric acid (HF) is also used to inscribe words or for abrasion on the glass. Glass forms soluble silicates with HF, hence HF is not kept in glass pots.

Note : A **carat** is a unit of mass equals to 200 mg (0.2 gram) which is used to weigh the diamond.

Buckminsterfullerene :

- It is a type of fullerene with the formula C_{60}.
- It has a cage-like fused-ring structure that resembles a **soccer ball,** made of **20 hexagons** and **12 pentagons,** with a carbon atom at each vertex of each polygon, and a bond along each polygon edge.
- It is used for drug delivery system in the body. They can act as hollow cages to trap other molecules. This is how they can carry drug molecules around the body and deliver them to where they are required and trap dangerous substances in the body and remove them.
- It is also used as a lubricant and catalyst.
- The tube fullerene is called **nanotube**.

Solid carbon dioxide :

- Solid (frozen) carbon dioxide is known as **dry ice**, because it looks like ice and it melts (sublimes) straight from solid to gas, without changing in liquid. Dry ice sublimes at –78.5°C (–109.3°F) at earth atmospheric pressure.
- It is useful for preserving frozen foods where mechanical cooling is unavailable.
- It does not leave any residue.

Types of Coal :

- On the basis of the presence of carbon percentage, different varieties of coal mineral have been divided into 4 main types–

(i) Lignite – 25-35% carbon

(ii) Sub Bituminous – 35-45%

(iii) Bituminous – 45-85% carbon

(iv) Anthracite – more than 85% carbon

Question Bank

1. Which one of the following elements forms the maximum number of compounds?

(a) Hydrogen (b) Carbon

(c) Nitrogen (d) Oxygen

U.P. P.C.S. (Mains) 2016

Ans. (b)

Among the above elements, carbon forms the maximum number of compounds.

2. Which of the following do not consist carbon?

(a) Diamond (b) Graphite

(c) Coal (d) None of these

42nd B.P.S.C. (Pre) 1997

Ans. (d)

Diamond is an additional form (allotrope) of carbon. Its relative density is 3.5. Graphite is a slaty-black coloured smooth and brightly organic matter of relative density 2.25, which is also a allotrope of carbon, while coal is a solid carbon containing organic matter used as fuel. Thus, all of the three consist carbon.

3. Graphite and diamonds are basically made up of atoms of :

(a) Silica (b) Carbon

(c) Nitrogen (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (b)

See the explanation of above question.

4. Thermodynamically the most stable form of carbon is :

(a) Fullerenes (b) Diamond
(c) Coal (d) Graphite

U.P. P.C.S. (Pre) 2022

Ans. (d)

Thermodynamically the most stable form of carbon is graphite. Thermodynamically, graphite has a high melting point, modest thermal expansion, and high specific heat, making it relevant for composites and structural materials and as heat shields in certain applications, including nuclear reactors. Graphite consists of layers of carbon sheets (graphene sheets) that are only weakly bonded with each other causing great anisotropy in the material.

5. Which of the following statements about graphite and diamond is true?

(a) They have the same crystal structure.
(b) They have the same hardness.
(c) They have the same electrical conductivity.
(d) They can undergo the same chemical reaction.
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

Graphite and diamond are both composed of carbon (C), but physically, they are very different. The differing physical properties of graphite and diamond arise from their distinct crystal structures. In a diamond, the carbon atoms are arranged tetrahedrally. The carbon atoms in graphite are also arranged in an infinite array, but they are layered in trigonal planar. Another important physical difference is their hardness. The hardness of minerals is compared using the Mohs Hardness Scale, a relative scale numbered 1 (softest) to 10 (hardest). Graphite is very soft and has a hardness of 1 to 2 on this scale. Diamonds are the hardest known natural substance and have a hardness of 10. Diamond is an insulator and does not conduct electricity, while graphite is a good conductor of electricity. Despite extreme differences in physical properties, diamond and graphite are chemically the same and can undergo the same chemical reaction.

6. Which of the following statements about diamond and graphite is true?

(a) They have same electrical conductivity.
(b) They have same crystal structure.
(c) They have same degree of hardness.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (e)

See the explanation of above question.

7. In the context of lab-grown diamonds (LGDs), what is used as a diamond seed?

(a) White sapphire
(b) Moissanite
(c) Graphite
(d) Cubic zirconia (CZ)

69th B.P.S.C. (Pre) 2023

Ans. (c)

In the context of lab-grown diamonds, the seed is a small diamond fragment or a carbon source (mainly graphite). Diamond structure can be produced by graphite by applying very high temperature and pressure. Catalysts such as chromium, iron or platinum may be used. White sapphire is aluminium oxide. It cannot form diamond which is pure carbon. There are two main lab-grown diamond production methods : the high pressure / high temperature process (HPHT) and chemical vapour deposition (CVD). Both methods are commonly used, but CVD is becoming more popular for producing gem-quality synthetic diamonds for jewellery. HPHT is used more often to create synthetic diamonds for industrial use.

8. Which of the following seed is used in production of engineered diamonds?

(a) Moissanite (b) White sapphire
(c) Graphite (d) Cubic Zirconia

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

See the explanation of above question.

9. Arrange the following substances in chronological order of their first synthesis in lab :

1. Black gold **2. Fullerene**
3. Graphene **4. Kevlar**

Select correct answer from the codes given below :

Codes :

(a) 1 2 3 4 (b) 4 2 3 1

(c) 2 4 3 1 (d) 4 1 2 3

U.P.P.C.S. (Pre) 2019

Ans. (b)

Kevlar is a heat-resistant and strong synthetic fiber which was developed by Stephanie Kwolek at Dupont lab (USA) in 1965.

Fullerene was first discovered in 1985 by British scientist Harold Kroto and American scientists Richard E. Smalley, James R. Heath and Robert F. Curl Jr.

Graphene is an allotrope of carbon which was first isolated and synthesized in 2004 by Andre Geim and Konstantin Novoselov at the University of Manchester.

In 2019, Scientists at the Mumbai-based Tata Institute of Fundamental Research (TIFR) used gold nanoparticles and by rearranging size and gaps between them developed a new material **'black gold'**, which has unique properties such as capacity to absorb light and carbon dioxide.

10. Which of the following is made up of Carbon only?

(a) Kevlar (b) Lexan

(c) Graphene (d) Spider silk

U.P.P.C.S. (Mains) 2015

Ans. (c)

Graphene is an allotrope of Carbon in the form of a single layer of atoms in a two-dimensional hexagonal lattice in which one atom forms each vertex. Thus, it is a nano structure of carbon atoms. It is the basic structural element of other allotropes including Graphite, Charcoal, Carbon nanotubes and Fullerenes. Graphene is a form of Carbon nanostructure.

11. Graphene is

(a) An alloy of carbon

(b) Nano structure of carbon

(c) Isotope of carbon

(d) None of the above

U.P. P.C.S. (Pre) 2018

Ans. (b)

See the explanation of above question.

12. Which of these is not an additional form of Carbon?

(a) Diamond (b) Graphite

(c) Oxocarbon (d) Fullerenes

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (c)

Oxocarbon is not an additional form (allotrope) of carbon. Oxocarbon is an inorganic compound consisting only of carbon and oxygen.

13. Pencil lead is :

(a) Graphite (b) Charcoal

(c) Lamp black (d) Coal

Chhattisgarh P.C.S. (Pre) 2011

U.P.P.C.S. (Pre) 1994

Ans. (a)

Graphite and Diamond both are allotropes of Carbon. Pencil cores are made of Graphite mixed with a clay binder which leaves grey or black marks that can be easily erased.

14. Lead Pencil contains –

(a) Lead (b) Lead oxide

(c) Graphite (d) Lead sulphide

U.P.P.C.S. (Mains) 2006

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (c)

See the explanation of above question.

15. Consider the following statements :

1. Carbon fibres are used in the manufacture of components used in automobiles and aircrafts.

2. Carbon fibres once used cannot be recycled.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2023

Ans. (a)

Carbon fibers can be defined as fibers with a carbon content of 90% or above. They are produced by thermal conversion of organic fibers with a lower carbon content such as polyacrylonitrile (PAN) containing several thousand filaments with diameter between 5 and 10 μm. The main advantage of carbon fibers compared to other fibers are the high tensile strength, high stiffness, low density, and a high chemical resistance. Carbon fibers are usually combined with other materials to form a composite. Carbon fiber-reinforced composite materials are used to make aircraft and spacecraft parts, racing car bodies, golf club shafts, bicycle frames, fishing rods, automobile springs, sailboat masts, and many other components where light weight and high strength are needed. Hence, statement 1 is correct.

Carbon fiber (and carbon fiber composites) waste can be recycled using various mechanical, thermal, or chemical methods including virgin carbon fibre offcuts, carbon fibre-reinforced composites (CFRC), utilizing chemical reactions, size-reduction method using high-voltage, electrohydraulic and electrodynamic fragmentation, etc. Hence, statement 2 is not correct.

16. **Third allotrope of Carbon was discovered by three scientists, who were awarded the Nobel Prize for Chemistry. Find out who was not on that team–**
(a) H.W. Kroto (b) R.F. Curl
(c) R.E. Smalley (d) Faimen

U.P.P.C.S. (Pre) 2009

Ans. (d)

In 1996, Robert F. Curl (USA), Sir Herold W.Kroto (Britain) and Richard E. Smalley (USA) were awarded the Nobel Prize in Chemistry for the discovery of Fullerenes. It was the allotrope of Carbon in which the molecules of Carbon are fully condensed. Its molecular formula is C_{60}. It is named as Fullerenes after the name of the American architect and inventor Richard Buckminster Fuller.

17. **Which one of the following pairs is not correctly matched?**
(a) Dry ice : Solid carbon dioxide
(b) Sevin : Insecticide
(c) Teflon : Polymer containing fluorine
(d) Fullerene : Organic compound containing fluorine

U.P. P.C.S. (Mains) 2017

Ans. (d)

Fullerene is an allotrope of carbon. Buckminsterfullerene is the cage-like molecule of fullerene which is composed of 60 carbon atoms (C_{60}) joined together by a single and double bond and to form a hollow sphere with 12 pentagonal and 20 hexagonal faces - a design that resembles a Soccer ball. The C_{60} molecule was named after the American Architect Fuller. Other pairs are correctly matched.

18. **Buckminsterfullerene is –**
(a) A form of carbon compound of clusters of 60 carbon atoms bond together in polyhedral structure composed of pentagons or hexagons
(b) A polymer of fluorine
(c) An isotope of carbon heavier than C^{14}
(d) None of these

U.P.P.C.S. (Mains) 2010

Ans. (a)

See the explanation of above question.

19. **Which one of the following pairs is not correctly matched?**
(a) Fullerene – Organic compounds containing fluorine
(b) Dry Ice – Solid carbon dioxide
(c) Keratin – Protein found in the outer layer of human skin
(d) Mustard gas – Poisonous liquid used is chemical warfare

U.P.P.C.S. (Pre) 2001, 2003
U.P. Lower Sub. (Pre) 2003
U.P. Lower Sub. (Pre) 2002
U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (a)

Dry Ice – We can simply say that dry ice is solid carbon dioxide CO_2. It is used as a cooling agent.
Mustard gas – It is used as a strong chemical weapon. This mortally chemical affects skin, leering eye, lungs and D.N.A. which affects the cells most. After 1-6 hours the symptoms seem to be visible.
Fullerene – Fullerene is an allotrope of the carbon family in which fully carbon atoms are integrated. It is denoted by C_{60}.
Keratin – It is a family of fibrous structural proteins. Keratin is the protein that protects epithelial cells from damage or stress that has potential to fill the cell. It is the key structural material making up the outer layer of human skin.
Teflon – Its commercial name is Polytetrafluoroethylene. It has been registered in 1944. It is synthesized fluoropolymer in which fluorine atom is mixed. So Teflon is a fluorine containing polymer.

20. **Which one of the following pairs is not correctly matched?**
(a) Dry ice : Solid carbon dioxide
(b) Mustard gas : Poisonous liquid used in chemical warfare
(c) Teflon : Polymer containing fluorine
(d) Fullerene : Organic compounds containing fluorine

U.P.P.C.S. (Pre) 2006

Ans. (d)

See the explanation of above question.

21. **Which one of the following pairs is incorrectly matched?**
(a) Pyrene – Fire extinguisher
(b) Sulphur Dioxide – Acid rain
(c) Freon – Refrigerant
(d) Fullerene – Fluorine containing polymer

U.P. P.C.S. (Mains) 2016

Ans. (d)

A Fullerene is an allotrope of carbon in the form of a hollow sphere, ellipsoid, tube and many other shapes. Rest of the options are correctly matched.

22. **Which of the following do not contain carbon ?**
(a) Diamond (b) Graphite
(c) Coal (d) Sand
U.P. Lower Sub. (Mains) 2013

Ans. (d)

Carbon is capable of forming many allotropes due to its valency. The well-known forms of Carbon included Diamond and Graphite. Coal is a sedimentary organic rock that contains a lot of Carbon (between about 40% to 90% carbon by weight). Sand is a naturally occurring granular material composed of finely divided rock and mineral particles. It does not contain Carbon.

23. **Which of these consists carbon?**
(a) Lignite (b) Tin
(c) Silver (d) Iron
U.P.P.C.S. (Pre) 1993

Ans. (a)

Coal minerals are divided mainly into four types on the basis of the percentage of Carbon content –
Lignite – 25-35%
Sub-bituminous – 35-45%
Bituminous – 45-85%
Anthracite – more than 85%.

24. **Which one of the following types of coal contains a higher percentage of Carbon than the rest?**
(a) Bituminous coal (b) Lignite
(c) Peat (d) Anthracite
U.P.U.D.A./L.D.A. (Pre) 2001
I.A.S. (Pre) 1999

Ans. (d)

See the explanation of above question.

25. **The highest amount of Carbon is in –**
(a) Pig Iron (b) Wrought Iron
(c) Steel (d) Alloy Steel
U.P.P.C.S. (Mains) 2014

Ans. (a)

Pig Iron has a very high Carbon content, typically 3.5–4.5%. Wrought Iron is an Iron alloy with a very low Carbon (less than 0.08%) content. Steels containing about 0.2% to 1.5% Carbon are generally known as Carbon steel. The amount of Carbon in alloy steel ranges from about 0.1% to 1%.

26. **Which of the following is not in the form of crystal?**
(a) Diamond (b) Quartz
(c) Sulphur (d) Graphite
M.P.P.C.S. (Pre) 1996

Ans. (c)

Salt, Sugar, Diamond, Quartz, Ice, Graphite etc. are in the form of crystal but Sulphur is not. It is a multivalent non-metallic chemical element.

27. **Consider the following statements :**
Glass can be etched or scratched by –
1. Diamond
2. Hydrofluoric Acid
3. Aquaregia
4. Conc. Sulphuric Acid
Which of these statements are correct ?
(a) 1 and 4 (b) 2 and 3
(c) 1 and 2 (d) 2 and 4
I.A.S. (Pre) 1999

Ans. (c)

Diamonds are used for grinding, cutting, drilling and polishing. It is also used as an abrasive. Very small pieces of diamonds are embedded into grinding wheels, saw blades or drill bits. Diamond is used to cut or scratch the Glass. Hydrofluoric (HF) acid is one of the strongest inorganic acids which is used mainly for industrial purposes (e.g. glass etching, metal cleaning, electronics manufacturing).

28. **For the ceiling of diamonds, the unit of weight is carat. One carat is equal to –**
(a) 100 mg (b) 200 mg
(c) 300 mg (d) 400 mg
U.P.P.C.S. (Mains) 2013

Ans. (b)

The carat (ct) is a unit of mass equal to 200 mg and is used for measuring gemstones and pearls.

29. **Graphene is frequently in news recently. What is its importance?**
1. It is a two-dimensional material and has good electrical conductivity.
2. It is one of the thinnest but strongest materials tested so far.
3. It is entirely made of silicon and has high optical transparency.
4. It can be used as 'conducting electrodes' required for touchscreens, LCDs and organic LEDs.
Which of the statement(s) given above is/are correct?
(a) 1 and 2 (b) 3 and 4
(c) 1, 2 and 4 (d) 1, 2, 3 and 4
I.A.S. (Pre) 2012

Ans. (c)

Graphene is an allotrope of Carbon consisting of a single layer of atoms arranged in a honeycomb nanostructure. It is an extremely good electrical conductor and is highly resistant to heat and acids. The two scientists of Manchester University-Andre Geim and Konstantin Novoselov properly isolated and characterized it in 2004. The both were awarded Nobel Prize in Physics in 2010 for groundbreaking experiments regarding the two-dimensional material graphene. They extracted Graphene from a piece of Graphite. Graphene is a bi-dimensional element and due to this quality, controlling electricity is easier in it than a tri-dimensional element. Graphene is not only very thin but also strongest among all the materials tested so far. It is almost transparent, despite, being so dense that the smallest gas molecule cannot pass through it. It can be used as 'conducting electrodes' required for touchscreens, LCDs and organic LEDs. Graphene can be used to make faster computer chips, communication devices etc.

30. Which one of the following materials is strongest?

(a) German Silver (b) Brass
(c) Steel (d) Graphene

U.P.P.C.S. (Mains) 2015

Ans. (d)

See the explanation of above question.

31. Consider the following statements and select the correct answer from the codes given below –

(1) Water becomes harder due to the presence of Calcium Sulphate and it is not usable.
(2) Diamond is harder than copper and iron.
(3) Oxygen is the main component of air.
(4) Nitrogen is used in the manufacture of vegetable ghee.

Code :

(a) 1 and 2 (b) 3 and 4
(c) 1 and 3 (d) 2 and 4

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (a)

Hard water has high mineral content caused by the dissolved Magnesium Sulphate or Calcium Sulphate. This is due to the permanent hardness of sulphate salts which do not decompose on heating. Diamond is the hardest substance. By volume, dry air contains about 78% of Nitrogen. Vanaspati ghee, a fully or partially hydrogenated vegetable cooking oil is often used as a cheaper substitute for ghee. So, only statement 1 and 2 are correct.

32. Which lubricant is used for heavy machines?

(a) Bauxite (b) Phosphorus
(c) Graphite (d) Silicon oil

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

Graphite lubricant is a thin black, anti-seize and anti-friction carbonic substance. It is an allotrope of Carbon. Graphite grease is used to lubricate industrial and automotive machinery.

33. Which of the following is also used as a lubricant?

(a) Cuprite (b) Graphite
(c) Haematite (d) Cryolite

M.P.P.C.S. (Pre) 2017

Ans. (b)

Graphite is the pure crystalline allotrope of carbon. It is also used as lubricant. In its crystals, the carbon atoms are arranged in regular hexagons in layers. Graphite is soft and lubricant due to the presence of weak Vander walls force between its layers.

34. Dry ice is –

(a) Solid water (b) Mountain ice
(c) Solid CO_2 (d) Solid carbon mono oxide

Uttarakhand P.C.S. (Pre) 2002
44th B.P.S.C. (Pre) 2000
43rd B.P.S.C. (Pre) 1999
42nd B.P.S.C. (Pre) 1998

Ans. (c)

Dry ice sometimes referred to as solid CO_2 or dry snow is the solid (frozen) form of carbon dioxide.

35. Which one of the following is called 'dry ice'?

(a) Dehydrated ice (b) Solid hydrogen peroxide
(c) Solid water (d) Solid carbon dioxide

U.P.P.C.S. (Mains) 2014

Ans. (d)

See the explanation of above question.

36. Chemically Dry Ice is –

(a) Solid sulphur
(b) Ice made from distilled water
(c) Mixture of ice and common salt
(d) Solid carbon dioxide

Uttarakhand U.D.A./L.D.A. (Mains) 2006
U.P.P.C.S. (Mains) 2009
Jharkhand P.C.S. (Pre) 2010

Ans. (d)

See the explanation of above question.